

SunHyuk Hwang

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RESEARCH INTERESTS

I work on the security of satellite systems, with a current focus on the RF and communication-protocol layer — signal reception, protocol analysis, and exploitation. A through-line in my work is building open tools and simulators that make this research hands-on and reproducible without specialized hardware. I aim to extend this from the RF link into the software that runs spacecraft — onboard flight and ground-segment systems.

EDUCATION

M.S., Cybersecurity — Kyung Hee University

Mar 2025 – Feb 2027 (expected)

Seoul, Republic of Korea

- Graduate researcher, PWNLAB cybersecurity research lab (2025 – present).
- Research focus: satellite communication security.
- GPA: 4.17 / 4.3 (in progress).
- Advisor: Prof. Dae-Hee Jang.
- Thesis: *A Study on the Design of a Virtual Satellite Antenna (VSA) System for Satellite RF Hacking Research* (TBD).

B.S., Computer Engineering — Kyung Hee University

Mar 2019 – Feb 2025

Seoul, Republic of Korea

- Undergraduate researcher, PWNLAB cybersecurity research lab (2023 – 2024).
- Research focus: drone security.

PUBLICATIONS

1. **S. Hwang** et al., “Design and Implementation of a Virtual Satellite Antenna System for Satellite RF Hacking Research,” *Conference on Information Security and Cryptography — Summer (CISC-S’26)*, Busan, Korea, May 2026. Oral presentation. [Domestic]
2. **S. Hwang** et al., “Analysis of Security Threats in Satellite Control Communication Using the NOS3 Satellite System Simulator,” *Conference on Information Security and Cryptography (CISC-W’24)*, Nov 2024. [paper] [Domestic]
3. **S. Hwang** et al., “Research on Exploiting Open-Source Ground Stations through Satellite Spoofing Signal,” *Korea Computer Congress (KCC 2024)*, Jun 2024. [paper] [Domestic]
4. **S. Hwang** et al., “Analysis of the Communication Protocol in Open-Source Drone (PX4),” *Conference on Information Security and Cryptography — Winter (CISC-W’23)*, Dec 2023. [Domestic]

RESEARCH EXPERIENCE

Graduate Researcher — Satellite Cybersecurity

Mar 2025 – present

PWNLAB, Kyung Hee University

- Research on the RF and protocol layer of satellite security — operating real and virtualized ground stations to study signal reception, tracking, and exploitation.
- Creator of [OpenVSA](#) and [Mr.Radio](#) (detailed under Selected Projects) — open tools that make satellite RF security research and education hands-on.

Ground System Lead (RLE) — KARI-hosted CubeSat competition

Jun 2026 – present

KHU team · 7th Korean CubeSat Competition (KASA / KARI)

- Lead the ground-system software segment; designing the Mission Operations Center (MOC) that commands the satellite and processes its telemetry and mission data.

Researcher — Satellite Cybersecurity R&D Project

2024 – 2025

Korean government research institutes

- Government-funded defense satellite security research.
- Technical details and deliverables restricted under NDA.

TEACHING EXPERIENCE

Teaching Assistant — Information Security (Undergraduate)

Jun 2026

Kyung Hee University · Instructor: Prof. Dae-Hee Jang

- Led two hands-on RF-hacking lab sessions covering RF attack fundamentals and practical signal exploitation.

RF Security Instructor — PWNLAB

Mar 2024 – present

PWNLAB, Kyung Hee University

- Designed and deliver internal RF-security training for lab members, introducing RF attack surfaces and hands-on exploitation techniques.

TALKS & DEMONSTRATIONS

• DEF CON 2026 Aerospace Village

Aug 2026

Co-developed and present hands-on satellite-hacking scenarios built on OpenVSA, in collaboration with IERAE, for visitors at the Aerospace Village, Las Vegas, NV, USA.

- **Invited talk** — RF security research overview, Korean government research institute, Daejeon, Korea Feb 2024

SELECTED PROJECTS

VSA / OpenVSA — Virtual Satellite Antenna

github.com/whal-e3/OpenVSA

- Desktop application that virtualizes a full satellite ground station — GPredict integration for real-time tracking, IQ signal-file analysis, and multi-antenna beam-pattern simulation — enabling hands-on satellite RF security research without physical hardware.

Mr.Radio — RF-Hacking Wargame Platform

mrradio.kr

- Browser-based RF-hacking wargame with virtualized satellite signal environments — signal reception, demodulation, and decoding without physical SDR hardware. Built for satellite RF security education; over 100 registered users.

Hack-the-Drone CTF — RF Challenges & Operations

2023, 2024

hackthedrone.org

- Sole designer and author of all RF-domain challenges — and a drone flight-system challenge — across both years.

HONORS & AWARDS

- **AUTOCRYPT Special Award** — Automotive Hacking Competition (AutoHack).
- **Hacktaeon CTF Finalist** — national CTF hosted by Sejong City.

TECHNICAL SKILLS

- **Software-defined radio:** full-stack satellite RF work — from installing and aiming antennas to receive live signals, through SDR hardware (HackRF One, USRP), to analysis software (GNU Radio, Inspectrum) for demodulation and decoding.
- **Satellite systems:** satellite communication protocols, the NOS3 simulator, NASA's core Flight System (cFS), the TinyGS ground-station stack, and the CCSDS standard protocol.
- **Systems hacking:** reverse engineering and binary exploitation (IDA Pro).

References available upon request.